KAMALESH G

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SUMMARY

Computer Science student with a foundation in full-stack web development and exposure to machine learning and AI systems. Have built and deployed projects across both domains — from MERN applications to RAG-based tools. Looking for opportunities where I can contribute meaningfully while continuing to grow as an engineer.

EDUCATION

Bachelor of Engineering in Computer Science and Engineering *August 2023 - May 2027*
PSG College of Technology, Coimbatore

Class XII *2023*
Sri Gurukulam Senior Higher Secondary School, Hosur

Class X *2021*
Sri Gurukulam Senior Higher Secondary School, Hosur

SKILLS

Programming Languages: Java, Python
Web Development: HTML, CSS, JavaScript, Node.js, React.js
Machine Learning: Scikit-learn, Pandas, NumPy
Databases: MongoDB, Vector Database (Qdrant), MySQL
Tools & Version Control: Git, GitHub

PROJECTS

Community Connect *MERN Application*

- Developing a full-stack platform to connect volunteers with NGOs based on interests, location, and opportunities.
- Implemented user authentication, NGO dashboards, and volunteer onboarding workflows.
- Built backend APIs using Node.js and MongoDB for managing users, events, and interactions.

Research Paper Summarizer *RAG Multi-Agent System*

- Built an end-to-end system to analyze and summarize research papers using a Retrieval-Augmented Generation (RAG) pipeline.
- Developed backend services using FastAPI for document processing, querying, and response generation.
- Designed an interactive interface using Streamlit for file upload, visualization, and real-time querying.
- Engineered and benchmarked the RAG workflow with real-time performance logging, achieving semantic retrieval latency as low as 1.2s and end-to-end response generation under 15s.
- Integrated LLM-based features to provide context-aware summaries and question-answering from documents.

WellNest *Web Application*

- Built a web platform for booking and conducting online counselling sessions with real-time video integration using Jitsi.
- Developed backend services using Node.js and MongoDB to manage users, sessions, and scheduling.
- Integrated EmailJS for automated notifications and reminders for appointments.

Diamond Price Prediction *Machine Learning*

- Built a machine learning regression model to predict diamond prices using features such as carat, cut, color, and clarity, achieving an R^2 score of 0.886 with Linear, Ridge, and Lasso Regression models.
- Performed data preprocessing, feature engineering, and exploratory analysis to improve model performance.
- Trained and evaluated models including Linear, Ridge, and Lasso Regression using RMSE and R^2 metrics.
- Deployed the model using a Streamlit web application for real-time predictions.

COURSEWORK

Data Structures and Algorithms, Object-Oriented Programming, Machine Learning, Design and Analysis of Algorithms, Database Management Systems

CERTIFICATIONS AND EVENTS

Earned a certificate in Responsive Web Design from freeCodeCamp.
Completed JavaScript Bootcamp by LEARNZ Development Hub.

CLUBS AND MEMBERSHIPS

Designing Head at PSG RadioHub — created visual content and event designs.
Member of IEEE–PSG Tech — actively involved in technical events.